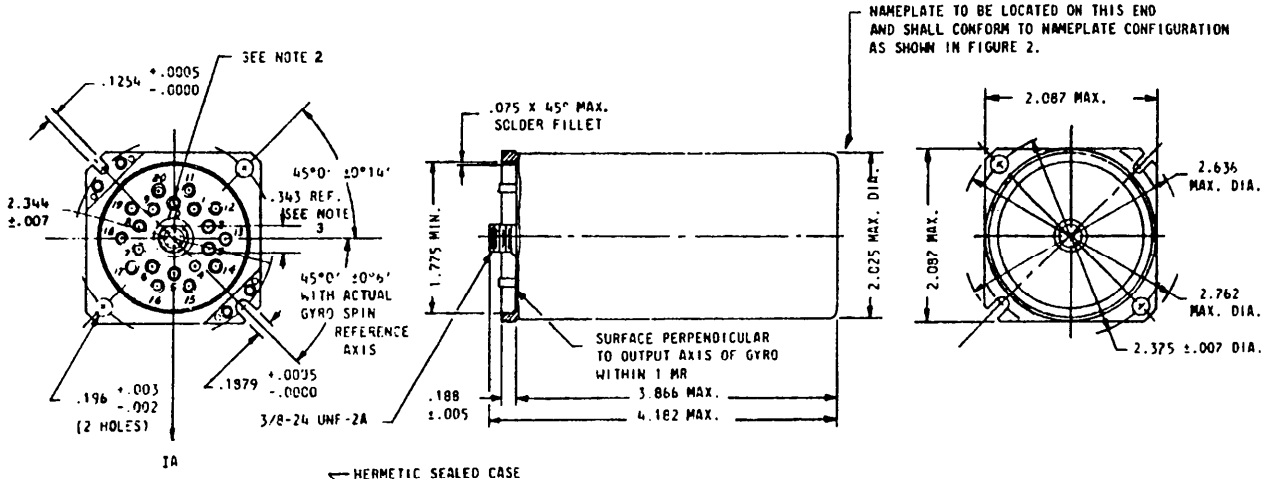
MIL-G-81168/3A(OS)
19 April 1967
SUPERSEDING
MIL-G-81168/3(WP)
5 February 1965
OS 8603
15 May 1964
BUORD DWG 1473417 (A)

# MILITARY SPECIFICATION SHEET
# GYROSCOPE, RATE INTEGRATING

THE COMPLETE REQUIREMENTS FOR PROCURING THE GYRO DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-G-81168. IN THE EVENT OF CONFLICT BETWEEN THIS DOCUMENT AND ANY OF THE DOCUMENTS FORMING A PART OF THIS SPECIFICATION, THE REQUIREMENTS OF THIS SPECIFICATION SHALL GOVERN.



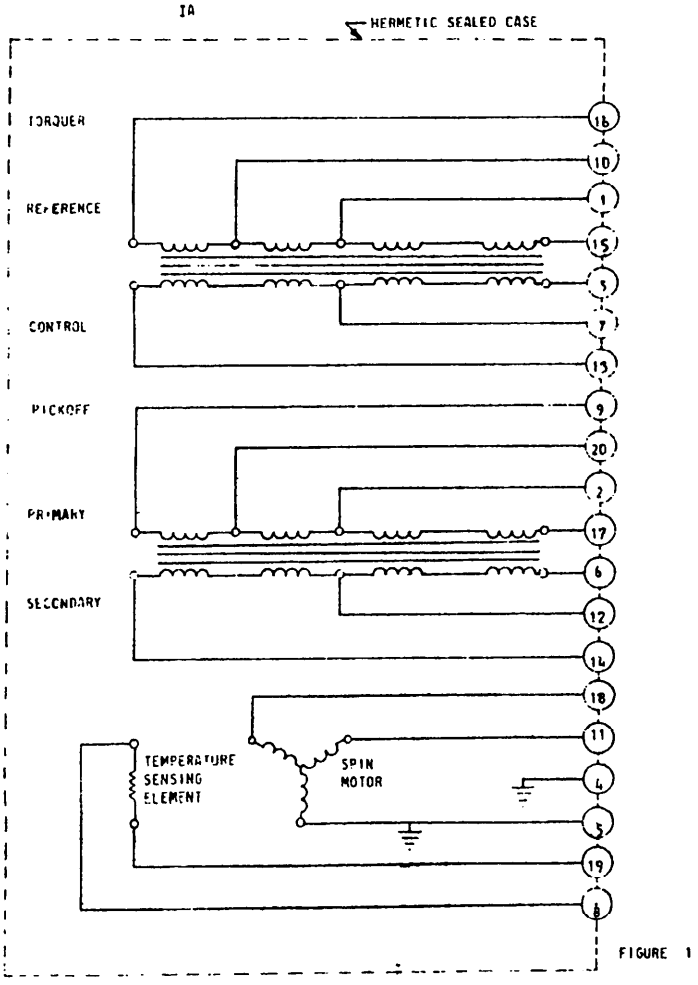
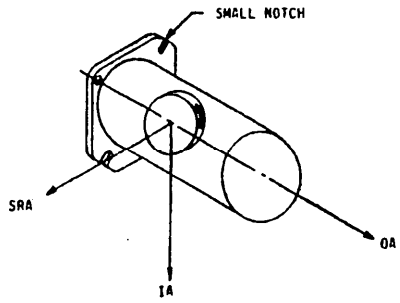




FIGURE 1




NOTES:

1. CONNECTOR TERMINALS 3, 14, 16 AND 17 SHOULD BE GROUNDED FOR MINIMUM MICROSYN CAPACITANCE EFFECTS AT GYRO NULL.
2. 20 CONTACT PINS 0.090 DIA. 0.22 LONG 10 PINS EQUALLY SPACED ON 0.813 DIA. BC AND 10 PINS EQUALLY SPACED ON 1.218 DIA. BC 18° BASIC FROM PINS ON 0.813 DIA. BC
3. INDEX FLAT PARALLEL TO CENTERLINE AND THRU PINS 13 AND 18 WITHIN ±1°.
4. THE OUTLINE CONFIGURATION IS FOR REFERENCE ONLY.
5. THE GYRO SHALL BE WARRANTEED FOR 2000 HOURS OF SPIN MOTOR OPERATION AT OPERATING TEMPERATURE, OR FOR ONE YEAR, WHICHEVER OCCURS FIRST.
6. THE GYRO SHALL HAVE A MINIMUM SHELF LIFE OF 3 YEARS AT AMBIENT TEMPERATURES WITHIN THE SPECIFIED LIMITS.
7. TEST MOUNTING ACCURACY: AXIS ALIGNMENT TESTS ± 0.5 ARC MINUTES. ALL OTHER TESTS ± 0.5 DEGREES.

(A)

FSC-1210

(A) For changes see pages 1 and 4.

REVISED

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Dielectric Strength: 500 volts rms, 60 cps at the following test points:

17 to case
6 to case
17 to 6
15 to case
3 to case
15 to 3
8 to case

Insulation Resistance: Terminal test points same as for Dielectric Strength. Test potential 500 volts d.c.

Spinmotor

| | | | |
|---|---|---|---|
| Type: | synchronous, hysteresis | | |
| Winding Configuration: | 3 phase | | |
| Rated Excitation: | | | |
| start | 10.5 | volts | ±1.0 |
| run | 9.0 | volts | ±1.5 |
| | Spinmotor shall run at synchronous speed from 380 to 420 cps. | | |
| Test Excitation:* | 10.0 | volts | ±0.5 |
| | 400 | cps | ±0.1% |
| | 2 | % distortion | maximum |
| Angular Velocity: | 12,000 | rpm | nominal |
| Rotor Angular Momentum: | $2.0 \times 10^4$ | gm-cm$^2$/sec | nominal |
| Run-up Time: | 15 | sec | maximum |
| Power Consumption: | 3.0 | watts | maximum |
| | 0.5 | power factor | ±0.1 |

Pickoff

| | | | |
|---|---|---|---|
| Type: | four pole, variable reluctance microsyn | | |
| Rated Excitation: | 0-100 | ma | range |
| Test Excitation:* | 100 | ma | ±0.5% |
| | 400 | cps | ±0.1% |
| | 2 | % distortion | maximum |
| Impedance: | | | |
| primary | 10.3 + j5.0 | ohms | ±12% |
| | with open circuit secondary | | |
| secondary | 2170 + j4520 | ohms | ±12% |
| | with open circuit primary | | |
| Null Voltage: | 10 | mv rms | maximum |
| Phase Shift: | 67 | degrees | ±5 |
| Scale Factor: | 26.5 | mv/mr | nominal |
| Linearity Error: | 5 | % | maximum. |
| | measured at input angles of 0, ±1, ±2, and ±3 degrees. | | |

For changes see pages 1 and 4.

*Note: Spinmotor, pickoff and torquer excitations shall be supplied from the same frequency source.



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## Torquer

| | | | |
|---|---|---|---|
| Type: | four pole, variable reluctance microsyn | | |
| Reference Winding Excitation: | 0-145 | ma | range |
| Control Winding Excitation: | 0-145 | ma | range |
| Test Excitation:* | 100 | ma | ±1% |
| | 400 | cps | ±0.1% |
| | 2 | % distortion | maximum |
| Command Rate Scale Factor: | 0.1000 | mr/sec/ma$^2$ | ±5% |
| | based on ± inputs to 5000 ma$^2$ inclusive. | | |

Torquer Linearity Error:

| ±Input ma$^2$ | Deviation mr/sec | Linearity Error percent |
|---|---|---|
| 250 | .45 | .020 |
| 750 | .85 | .038 |
| 2000 | 1.85 | .082 |
| 5000 | 4.25 | .189 |
| 10000 | 10.25 | .455 |

| | | | |
|---|---|---|---|
| Maximum Command Rate: | 2250 | mr/sec | nominal |
| Impedance: | | | |
| reference winding | 136 + j284 | ohms | ±12% |
| | with open circuit control winding | | |
| control winding | 196 + j302 | ohms | ±12% |
| | with open circuit reference winding | | |

## Temperature Sensor

| | | | |
|---|---|---|---|
| Type: | resistance alloy wire | | |
| Scale Factor: | 1.44 | ohms/°F | ±15% |
| Operating Temperature Resistance: | 750 | ohms | ±20 |
| Induced Voltage: | 3.0 | mv | maximum |
| Test Excitation: | 10 | ma | maximum |
| | 2 | % distortion | maximum |
| Operating Temperature: | 190 | °F | ±1 |

## Gyro Unit

| | | | |
|---|---|---|---|
| Gyro Scale Factor: | 30 | mv/mr | ±15% |
| Input Angular Freedom: | 7.0 | deg | ±0.5 |
| | from pickoff null | | |

Gyro phasing:

Spinmotor: With an ABC phase sequence, positive spinmotor rotation will result with terminal 18 connected to line A, terminal 5 connected to line B, and terminal 11 connected to line C.

Pickoff: Positive output voltage exists when voltage from terminal 14 to terminal 6 leads the voltage from 17 to terminal 9 by less than 90 degrees.

Torquer: Positive torque is obtained when terminals 3 and 16 are grounded and the currents to terminals 13 and 15 are in phase.

"(A) For changes see pages 1 and 4"



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| | | | |
|---|---|---|---|
| Characteristic Time: | 3.0 | ms | maximum |
| Axis Alignment: | OA orthogonal to mounting flange surface within 1 milliradian.<br>IA located 45°0' +0°6' clockwise (viewed from gyro terminal end) from the centerline of the mounting flange notches.<br>SRA located 45°0' +0°6' counterclockwise (viewed from gyro terminal end) from the centerline of the mounting flange notches. | | |
| Acceleration Insensitive Drift: | 975 | deg/hr | maximum |
| test procedure | 4.27 of MIL-G-81168 | | |
| Acceleration Sensitive Drift: | | | |
| mass unbalance | 40 | deg/hr | max. resultant |
| test procedure | 4.27 of MIL-G-81168 | | |
| anisoelasticity | 1.0 | deg/hr/g$^2$ | maximum |
| test procedure | 4.31 of MIL-G-81168 | | |
| Day-to-Day Stability | 20 | deg/hr | maximum |
| Ambient Temperature | -65 to +190 | degrees F | range |
| Operating Life: | 2000 | hrs | minimum |

Vibration, shock, altitude and moisture resistance shall be conducted in accordance with MIL-G-81168

Nameplate: Gyros shall be identified with markings conforming to MIL-STD-130. Markings shall be applied by acid, electric etching, permanent marking ink or engraving applied directly to the case. The markings shall be as permanent as the normal life expectancy of the gyro on which it is applied and legibility shall be as required for readability and identification. Markings shall withstand the same environmental tests required of the gyro.

Other markings on the gyro body shall include:

(1) Actual temperature sensor resistance ±1.25 ohms at gyro operating temperature and

(2) Actual command rate scale factor ±0.3 percent.

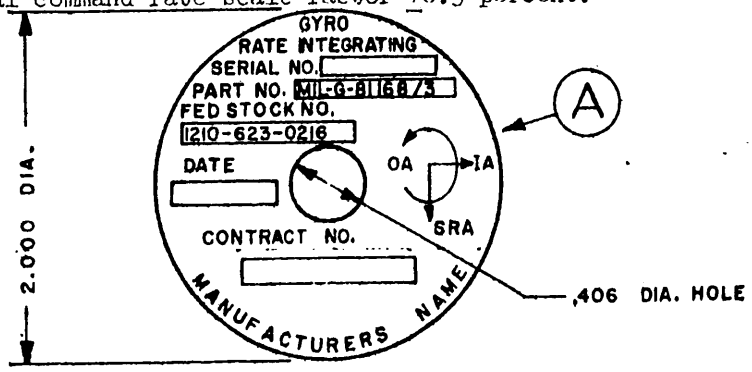



FIGURE 2

NAMEPLATE CONFIGURATION

Ⓐ For changes see pages 1 and 4"



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-G-81163/3A(OS) GYROSCOPE, RATE INTEGRATING

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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Naval Ordnance Systems Command
Washington, D.C. 20360

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DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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conducting first article approval tests, the contractor, prior to sub